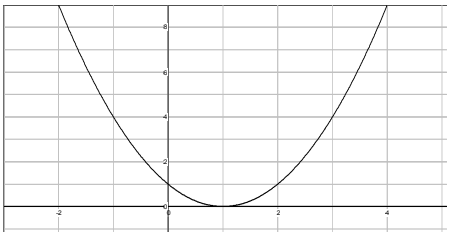
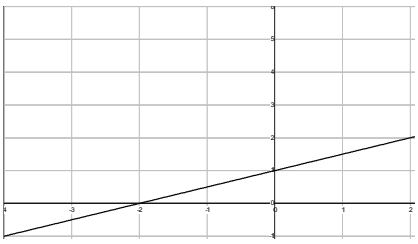
Question	Answer	Marks	Partial Marks
7(b)(iii)	0.28 oe	**3**	**M2** for $0.2 + 0.8 \times 0.1$ oe or **M1** for 0.2 or 0.8×0.1 or $0.8 \times (0.6 + 0.3)$
7(c)	0.32 oe	**3**	**M2** for $0.8 \times 0.2 + 0.2 \times 0.8$ oe **M1** for one of these products
8(a)(i)	36	**2**	**M1** for $\left(\frac{8}{12}\right)^2$ or $\left(\frac{12}{8}\right)^2$ oe
8(a)(ii)	30	**3**	**M2** for $320 \div 16 \times \frac{12}{8}$ oe or **M1** for $320 \div 16$
8(b)	3.375 cao	**3**	**M2** for $\frac{\frac{4}{3}\pi \times 4.5^3}{\pi \times 6^2}$ or better or **M1** for $\pi \times 6^2 \times h = \frac{4}{3} \times \pi \times 4.5^3$
8(c)	3.63 or 3.627 to 3.628	**3**	**M2** for $\frac{20^3}{40 \times \frac{4}{3}\pi}$ or **M1** for $40 \times \frac{4}{3} \times \pi \times r^3 = 20^3$
8(d)	$\frac{3x}{2}$ or $1.5x$ or $1\frac{1}{2}x$	**3**	**B2** for $4R^2 = 9x^2$ oe or better or **M1** for $4\pi R^2 = 2\pi x^2 + \pi \times 2x \times \frac{7x}{2}$
9(a)(i)	$(x+4)^2 - 25$	**2**	**B1** for $(x+k)^2 - 9 - (their\,k)^2$ or $(x+4)^2 - h$ or $k = 4$
9(a)(ii)	$x + 4 = [\pm]\ 5$	**M1**	**FT** *their* **(a)(i)**
	–9 and 1	**A1**	
9(b)	[b =] 7 [c =] –3	**3**	**B1** for [b =] 7 **M1** for $b^2 - 4c = 61$
9(c)(i)(a)	Correct sketch 	**2**	**B2** for correct quadratic curve with min touching x-axis or **B1** for parabola vertex downwards

Question	Answer	Marks	Partial Marks
9(c)(i)(b)	Correct sketch 	**2**	**B2** for correct straight line intersecting curve on y-axis or **B1** for straight line with positive gradient and positive y-intercept
9(c)(ii)	2.8[0] or 2.795...	**7**	**B3** for $x^2 - \frac{5}{2}x = 0$ oe or **M1** for $(x-1)^2 = \frac{1}{2}x + 1$ **B1** for [$(x-1)^2$ =] $x^2 - x - x + 1$ AND **B2** for (0, 1) and $\left(\frac{5}{2}, \frac{9}{4}\right)$ oe or **B1** [x =] 0 and $\frac{5}{2}$ oe AND **M1** for (difference in x)^2 + (difference in y)^2
10(a)(i)	5	**2**	**M1** for $(-1)^4 - 4(-1)^3$
10(a)(ii)	(0, 0) and (3, –27)	**6**	**B2** for $4x^3 - 12x^2$ [= 0] or **B1** for $4x^3$ or $12x^2$ AND **M1** for derivative = 0 or *their* derivative = 0 **M1** for $4x^2(x - 3)$[= 0] **B1** for [x =] 0 and [x =] 3 or [y =] 0 and [y =] –27 or for one correct coordinate pair
10(b)	[p =] 11 [q =] 5	**2**	**B1** for each or **M1** for $\frac{dy}{dx} = px^{p-1} + 2qx^{q-1}$